Synapse HR-ERP

Prediction models: previous vs current

Evaluation Report: Promotion Readiness, Performance Forecast and Attrition Risk Models

7 October 2026

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Summary

On 26 September 2026 the promotion readiness and performance forecast models in Synapse HR-ERP were rebuilt. This report compares the previous and current version of each model, and of the unchanged attrition risk model, on the same data and tests.

- **Performance forecast:** the current model is more accurate. It misses the next appraisal score by 4.64 points on average, against 6.44 for the previous model as it ran in the system.
- **Promotion readiness:** both versions rank employees equally well under real company conditions. Only the current model states true chances, ignores department names and personal attributes, and handles incomplete records correctly.
- **Attrition risk:** unchanged. It is better than guessing but not reliable enough to act on alone.
- The previous models scored higher only when given data the system does not provide.

Key results

Model	Measure	Previous, as run in the system	Current
Promotion readiness	Ranking (ROC-AUC), company department names	0.843	0.842
Promotion readiness	Gap between stated chance and real rate	12.4 points	0.7 points
Promotion readiness	Score depends on department name or personal attributes	Yes (department: 49.8 points on average)	No
Promotion readiness	Handles one appraisal, or none	No, fills in typical values	Yes
Performance forecast	Average miss (0 to 100 scale)	6.44 points	4.64 points
Performance forecast	Band correct (below, on track, exceeds)	71.4%	79.7%
Performance forecast	Forecast lowered by a better record	51 of 2,000 people	0 of 2,000
Performance forecast	Uses only facts known before the forecast	No	Yes
Performance forecast	Gives a likely range that holds	No range	Yes, 79.5% against a target of 80%
Attrition risk	Ranking (ROC-AUC)	0.628	0.628 (same model)
Attrition risk	Organisation that records no attendance	Fails with an error	Works
All three	Automated tests passed	49 (attrition only)	136 (all three models)

1. Method

Both versions were rebuilt from the project's version history in a separate workspace: the previous version as of 25 September 2026 and the current version as of 7 October 2026. Each was trained and tested on the same records, and the workspace was deleted afterwards.

Data

- **Promotion and performance:** a reference dataset of 100,000 employee records, 10% of them promoted. Both versions train on the same 80,000 records and are tested on the same 20,000 records, which neither saw in training.
- **Attrition:** the project's survey of 155 people, 50 of whom resigned.

Test conditions. The previous models expect about 40 facts per employee. The system records about 12 and filled the rest with typical values, so each previous model is tested in up to three conditions. The current models use only facts the system records and have one condition.

Condition	What the model is given
All 40 inputs	Every column of the dataset. Its developer tested it this way; the system never provided it.
As run, matching department names	The facts the system actually sent, the rest filled with typical values, and the dataset's own department names. This favours the previous model.
As run, company department names	The same, with department names that differ from the dataset's, as a real company's do.

Tests. Both versions were measured on ranking, decisions, calibration, error size, stability across data splits and statistical significance. Behaviour tests checked whether a better record ever lowers the result, whether department or personal attributes change it, how missing history and out-of-range values are handled, and whether the same input always gives the same output. Each version's own automated tests were also run.

Ranges in square brackets are 95% confidence intervals from 1,000 bootstrap resamples (random redraws of the test records). A difference is treated as real when its p-value, the chance of seeing it with no real effect, is below 0.05.

2. Promotion readiness

2.1 Ranking

ROC-AUC is the chance that the model scores a promoted employee above one who was not promoted (0.5 is guessing, 1.0 is perfect). PR-AUC measures how well the top of the ranking is filled with promoted employees (guessing scores 0.10). Five-fold cross-validation repeats training and testing on five different splits of the data.

Version and condition	ROC-AUC [95% range]	PR-AUC	Five-fold ROC-AUC (mean ± spread)
Previous, all 40 inputs	0.941 [0.936 to 0.944]	0.666	0.939 ± 0.003
Previous, as run, matching department names	0.923 [0.918 to 0.928]	0.607	0.923 ± 0.003
Previous, as run, company department names	0.843 [0.835 to 0.851]	0.373	Not run
Current	0.842 [0.834 to 0.850]	0.372	0.843 ± 0.004

With company department names, the previous model leads by 0.001 (95% range 0.0001 to 0.0025), which is too small to matter. With matching names it leads by 0.081 ($p < 0.001$), because department was its strongest input. In Figure 1, the previous model's curve falls onto the current model's once company names are used.

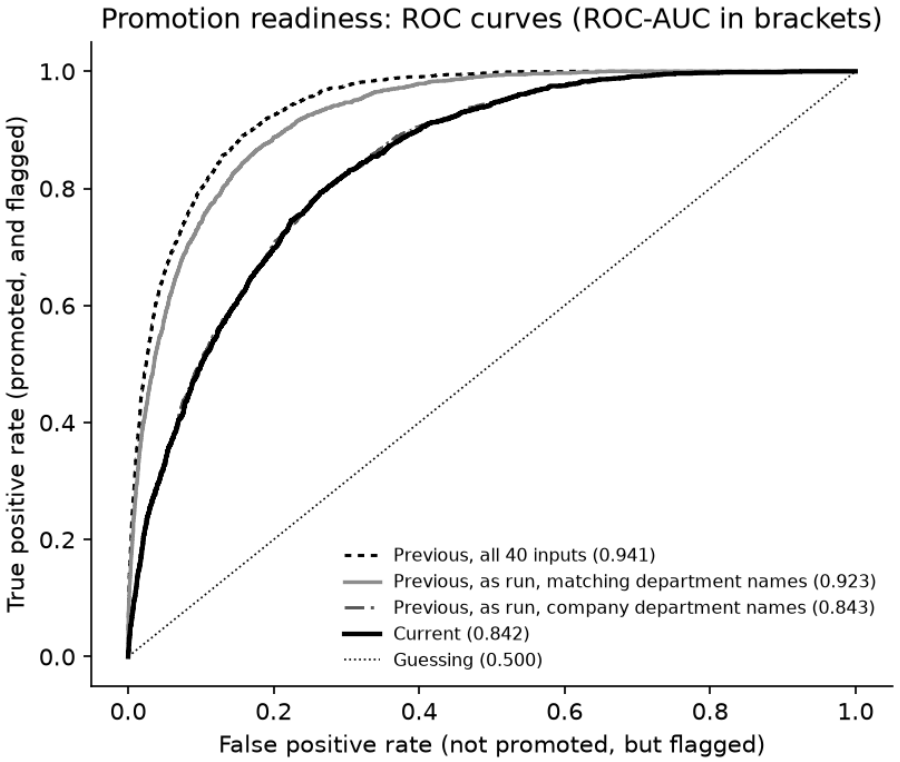


Figure 1. Promotion readiness ROC curves

2.2 Decisions

Each version flags its own *High* tier: a stated chance of 66% or more for the previous version, and 20% or more (twice the average rate) for the current one. Precision is the share of flagged employees who were promoted. Recall is the share of promoted employees who were flagged. F1 combines the two and is high only when both are high.

Version and condition	People flagged	Accuracy	Precision	Recall	F1
Previous, all 40 inputs	3,357	89.2%	47.5%	79.7%	59.5%

Version and condition	People flagged	Accuracy	Precision	Recall	F1
Previous, as run, matching department names	2,909	89.6%	48.5%	70.6%	57.5%
Previous, as run, company department names	2,252	87.5%	38.9%	43.9%	41.3%
Current	3,465	84.0%	32.8%	56.8%	41.6%

Answering “not promoted” for everyone is 90.0% accurate, so accuracy does not separate these models. Under real company conditions F1 is the same (41.3% and 41.6%), but the current model finds more of the employees who were promoted (56.8% against 43.9%).

Decision counts (20,000 test records; previous model as run with company department names)

Outcome	Current: flagged	Current: not flagged	Previous: flagged	Previous: not flagged
Promoted (2,000)	1,136 correct	864 missed	877 correct	1,123 missed
Not promoted (18,000)	2,329 wrongly flagged	15,671 correct	1,375 wrongly flagged	16,625 correct

2.3 Accuracy of stated chances

The calibration gap is the average distance, in percentage points, between the chance a model states and the rate that really occurs. The Brier score is the average squared error of the stated chances. Lower is better for both.

Version and condition	Average stated chance	Real rate	Calibration gap	Brier score
Previous, all 40 inputs	25.8%	10.0%	15.8 points	0.105
Previous, as run, matching department names	23.9%	10.0%	13.9 points	0.099
Previous, as run, company department names	22.4%	10.0%	12.4 points	0.113
Current	10.0%	10.0%	0.7 points	0.075

Stated chance and actual promotion rate by tier (20,000 test records)

Tier	Previous: stated chance	Previous (as run, matching names): promoted	Current: stated chance	Current: promoted
High	66% or more	48.5%	20% or more	32.8%
Medium	33% to 66%	14.2%	10% to 20%	15.2%
Low	Under 33%	1.5%	Under 10%	3.0%

The previous model was trained as if promoted and not promoted employees were equally common, so its percentages overstate the real rate more than twofold. Every current tier falls within its stated range. In Figure 2, the current model follows the diagonal of true chances; every previous condition falls below it.

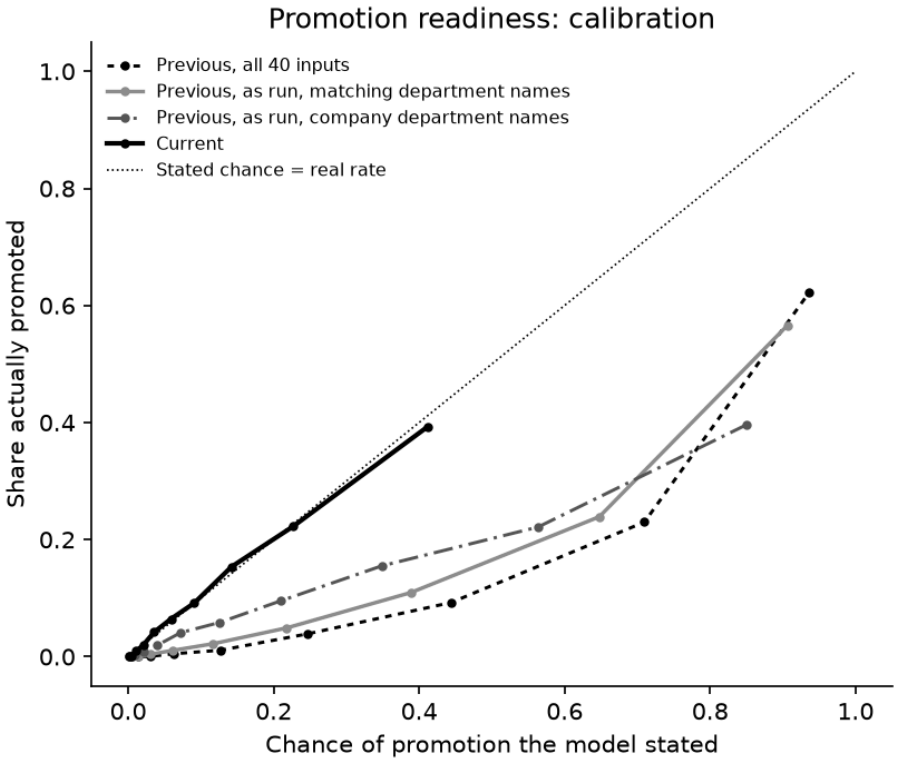


Figure 2. Promotion readiness calibration

2.4 Behaviour tests

Test	Previous	Current
Same employee moved through the seven departments	Stated chance changes by 49.8 points on average (up to 83.8); tier changes for 67.8% of people	No change (department is not an input)
Department name the model has not seen (“Information Technology”), typical employee	Stated chance rises from 19.7% to 49.8%	No change
Gender changed (all 40 inputs)	Changes by 1.5 points on average (up to 4.1); tier changes for 3.5%	No change (not an input)
Age raised 10 years, or marital status changed	Changes by under 0.1 points on average	No change (not inputs)
Only one appraisal on record	Missing scores filled with typical values: ROC-AUC 0.699, calibration gap 29.9 points (company department names)	Separate model for one appraisal: ROC-AUC 0.697, calibration gap 0.5 points
No appraisal on record	Still gives a score (8.4%), built entirely from filled-in values	Declined as “not enough data”
Latest score entered as 120 or 150	Stated chance near 100% (99.99999% at 120), no warning	Held at 100 with a warning

Test	Previous	Current
Latest appraisal raised 1 to 20 points	Never scores lower (0 of 2,000 people)	Never scores lower (0 of 2,000 people)
Same input twice, alone or in a batch	Identical	Identical

The previous model used gender, age, marital status, education and city as inputs. The current model uses only the latest appraisal score and its change since the previous appraisal.

Figure 3 varies the latest score for one typical employee (previous appraisal 68). The previous model approaches certainty from a score of about 85. The current model levels off near 69%, because even the strongest records in the data were not always promoted.

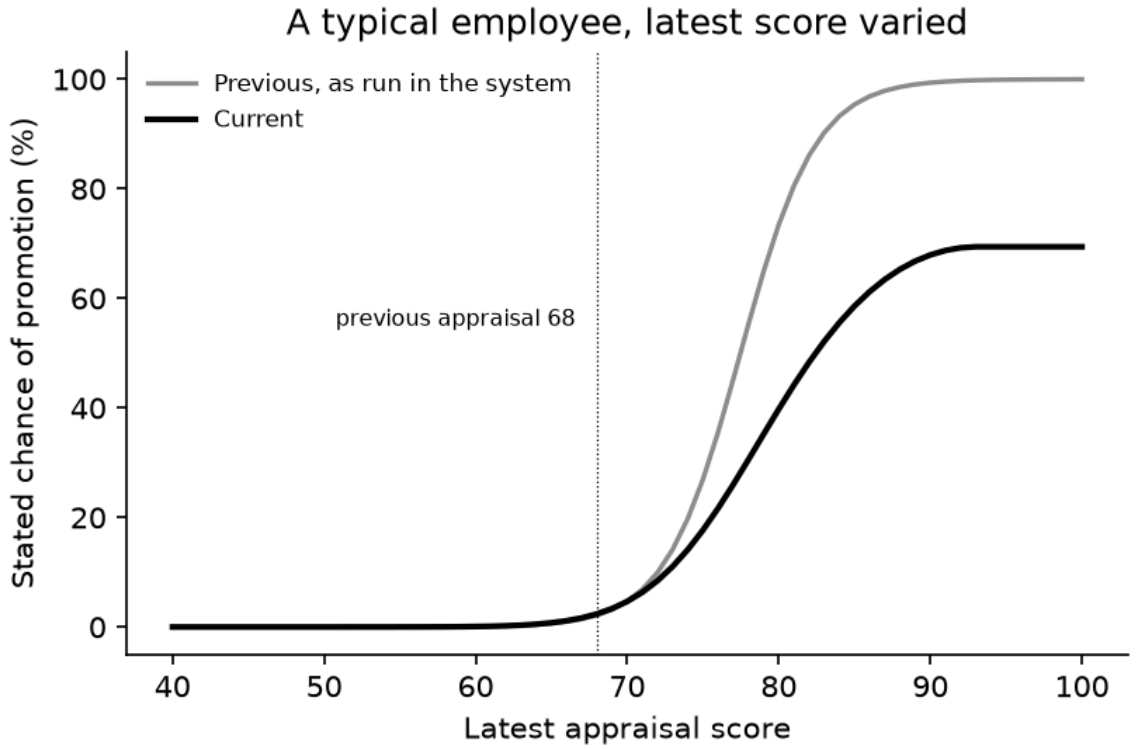


Figure 3. Promotion readiness response for a typical employee

3. Performance forecast

3.1 Accuracy

Average miss (MAE) is the mean distance between the forecast and the actual score. RMSE is similar but gives large misses more weight. R^2 is the share of the differences between employees' scores that the forecast explains. The bands are below (under 60), on track (60 up to 80) and exceeds (80 and over); precision, recall and F1 are averaged over the three bands.

Version and condition	Average miss [95% range]	RMSE	R^2	Band correct	Precision	Recall	F1
Previous, all 40 inputs (reads the forecast year)	3.33 [3.30 to 3.37]	4.18	92.0%	85.5%	86.0%	85.3%	85.6%
Previous, as run in the system	6.44 [6.38 to 6.50]	8.02	70.6%	71.4%	79.3%	66.3%	69.0%
Previous, system's facts placed correctly (generous)	7.06 [6.99 to 7.12]	8.58	66.3%	66.4%	85.2%	57.3%	59.9%
Current	4.64 [4.59 to 4.69]	5.82	84.5%	79.7%	80.5%	79.3%	79.8%
Simple rule: repeat the last score	4.86 [4.81 to 4.91]	6.15	82.7%	78.4%	80.0%	78.0%	78.5%
Simple rule: company average for everyone	12.10 [11.98 to 12.22]	14.78	0.0%	47.8%	15.9%	33.3%	21.6%

The current model misses by 1.80 points less than the previous model as it ran (95% range 1.74 to 1.86, $p < 0.001$). It placed 2,927 employees in the correct band that the previous model missed, against 1,257 the other way (McNemar test, $p < 0.001$). Five-fold cross-validation matched every average miss within 0.02 points.

The previous model's best result (3.33) cannot be reached in practice, because it reads the manager rating and KPI attainment of the year being forecast. Retrained without those and seven other same-year scores, the previous design misses by 4.16 points, but it still needs about 30 facts the system does not record.

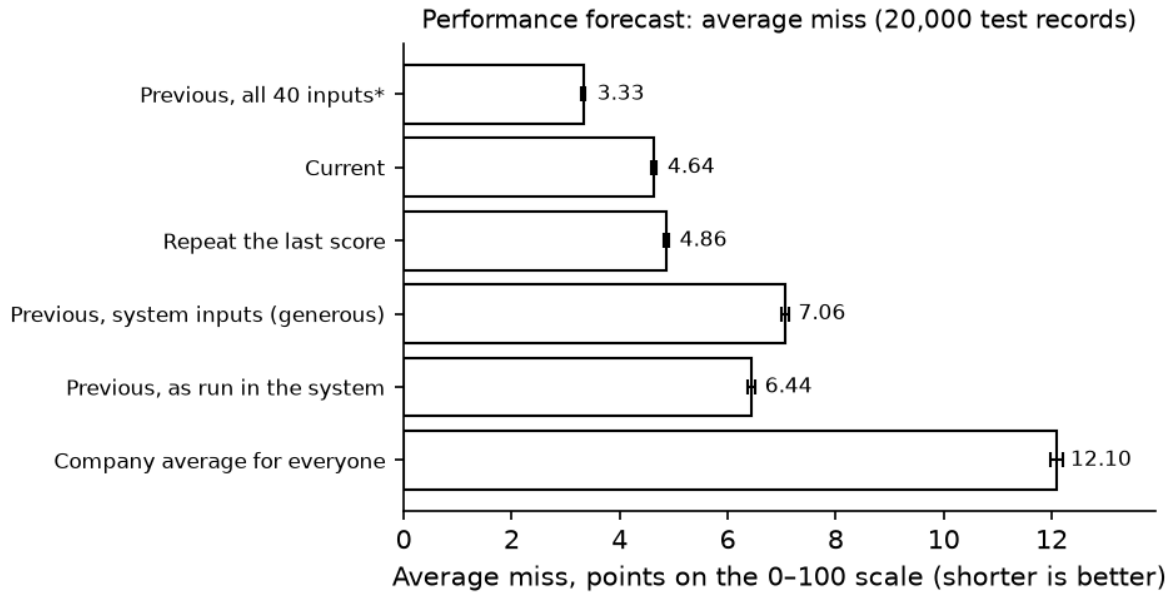


Figure 4. Performance forecast average miss (whiskers are 95% ranges)

3.2 Bands

Band counts (20,000 test records)

Actual band	Current model: predicted			Previous model, as run: predicted			Total
	Below	On track	Exceeds	Below	On track	Exceeds	
Below	4,062	1,071	0	3,264	1,869	0	5,133
On track	1,007	7,772	785	749	8,613	202	9,564
Exceeds	1	1,194	4,108	2	2,906	2,395	5,303

Band	Previous (as run): precision / recall / F1	Current: precision / recall / F1
Below	81.3% / 63.6% / 71.4%	80.1% / 79.1% / 79.6%
On track	64.3% / 90.1% / 75.1%	77.4% / 81.3% / 79.3%
Exceeds	92.2% / 45.2% / 60.6%	84.0% / 77.5% / 80.6%

The previous model pushed most employees toward *on track*. It identified 45.2% of those who went on to exceed, against 77.5% for the current model.

3.3 Range and confidence

The previous model gave a single number. The current model also gives a likely range and a confidence that the band is correct.

- **Range:** the actual score fell inside it for 79.5% of the 20,000 test records, against a target of 80%. It is 14.7 points wide on average.
- **Confidence:** stated confidence matches how often the band was correct, within 2 points at every level.

Stated confidence	Band actually correct
57%	59%
72%	70%
83%	83%
90%	89%
98%	98%

3.4 Behaviour tests

Test	Previous	Current
Latest appraisal raised 1 to 20 points	Forecast went down for 51 of 2,000 people (as run) and 57 of 2,000 (all inputs)	Never went down (0 of 2,000)
Facts from the year being forecast	Reads the manager rating and KPI attainment of that year	None

Test	Previous	Current
Only one appraisal on record	Average miss 9.17; band correct 47.8%, no better than answering “on track” for everyone	Needs only one: average miss 4.64; band correct 79.7%
No appraisal on record	Still forecasts (70.5 for a typical employee) from filled-in values	Declined as “not enough data”
Latest score entered as 120 or 150	Treated as 100, no warning	Held at 100 with a warning
Same employee in another department, or with gender changed	Under 0.01 points on average	No change (not inputs)
Same input twice, alone or in a batch	Identical	Identical

Figure 5 varies the latest score for one typical employee. The previous model, as run, moves only from 61 to 76 across the whole scale, because the system placed the latest score in the manager rating and KPI attainment inputs and the score before it in last year’s score. The current forecast follows the latest score; its 80% range is shaded.



Figure 5. Performance forecast response for a typical employee

4. Attrition risk

The model is the same in both versions and gives identical results. It ranks better than guessing (ROC-AUC 0.628, p = 0.03), but its accuracy is below that of answering “stays” for everyone. It should guide whom to talk to, not decide anything about a person. The only

change: the previous version fails with an error for an organisation that records no attendance, while the current version trains and scores all 155 respondents.

Measure (both versions)	Result
ROC-AUC, 10 repeats of five-fold cross-validation	0.628 ± 0.038
ROC-AUC, each person's scores averaged [95% range]	0.637 [0.547 to 0.731]
PR-AUC (guessing scores 0.323)	0.449
Brier score	0.224
Calibration gap	10.2 points
Trained on one survey group, tested on the other	ROC-AUC 0.358 and 0.392

Flag	People flagged	Accuracy	Precision	Recall	F1
Chance 50% or more	47	65.8%	46.8%	44.0%	45.4%
High tier (66% or more)	13	65.8%	38.5%	10.0%	15.9%
Medium or High (33% or more)	104	49.7%	36.5%	76.0%	49.4%
Answer "stays" for everyone	0	67.7%	Not defined	0.0%	Not defined

Decision counts (all 155 respondents, each scored by a model that never saw them; flagged = chance 50% or more)

Outcome	Flagged	Not flagged	Total
Resigned	22 correct	28 missed	50
Stayed	25 wrongly flagged	80 correct	105

Other methods on the same data scored lower (ROC-AUC): Extra Trees 0.626, Gradient Boosting 0.597 and Logistic Regression 0.543, against 0.628 for the Random Forest in use.

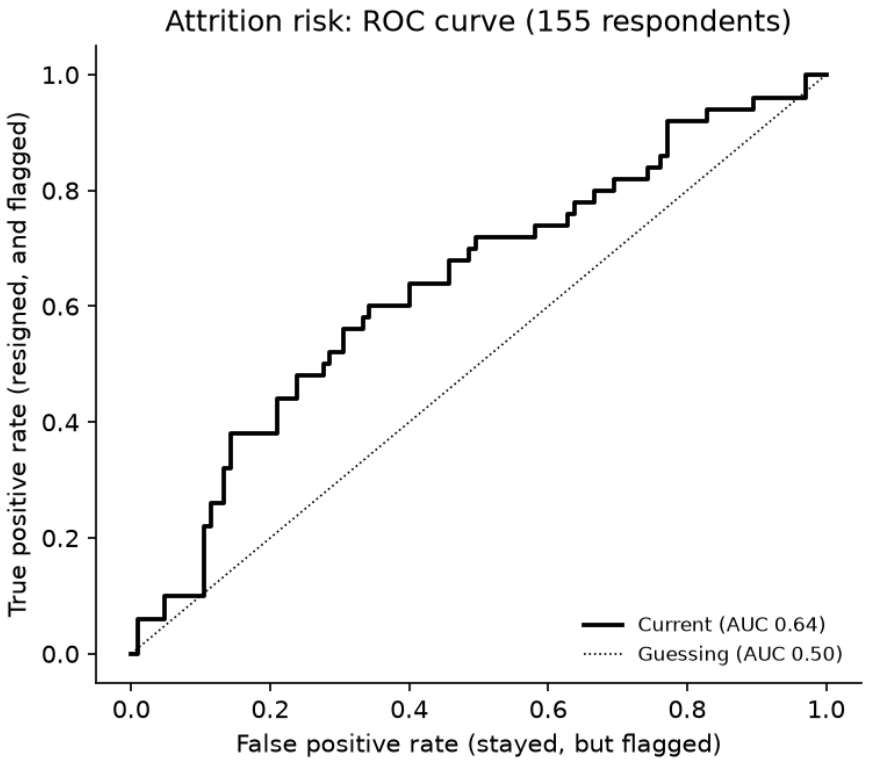


Figure 6. Attrition risk ROC curve

A permutation test checked whether the ranking is real. The outcomes were shuffled 100 times so that no real pattern remained. These runs averaged ROC-AUC 0.507, and only 3 of 100 reached the real result (Figure 7).

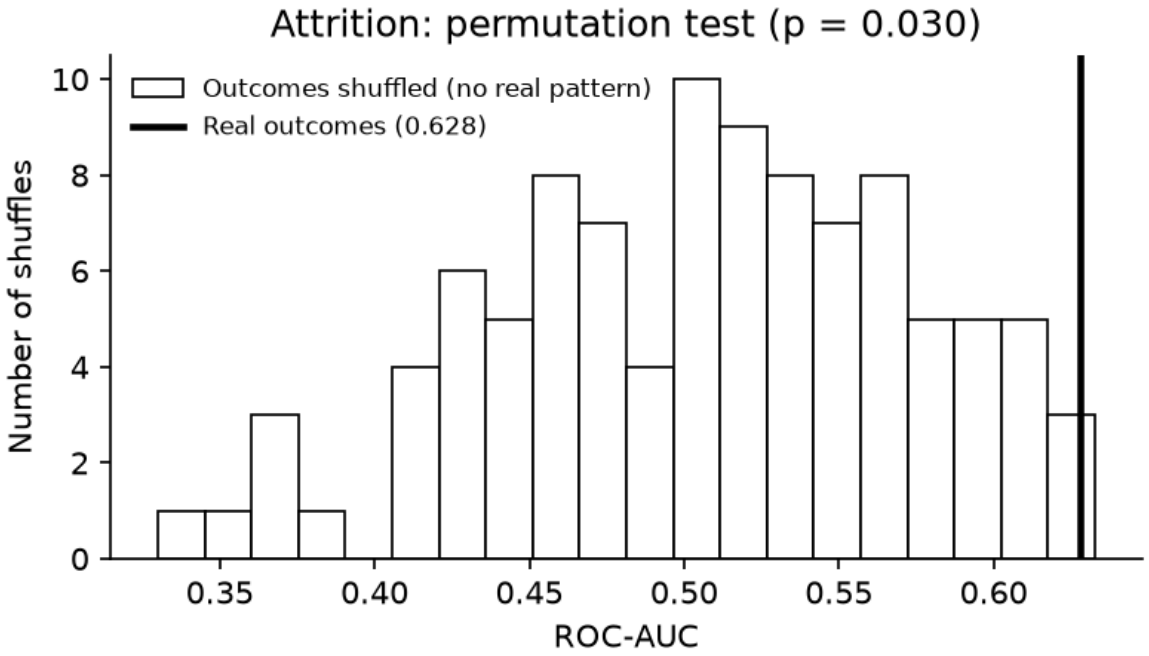


Figure 7. Attrition permutation test

5. Automated tests

Each version's own built-in tests were run on its own code against the trained models. All passed.

Area	Previous	Current
Attrition: survey cleaning	12	12
Attrition: inputs	22	22
Attrition: scoring service	15	15
Promotion model	None	18
Performance model	None	11
Shared appraisal handling	None	27
Saved promotion and performance models as served	None	12
Models trained on a company's own records	None	19
Total	**49 of 49**	**136 of 136**

6. Limitations

- **Reference data, not company data.** Promotion and performance were trained and tested on a reference dataset, not on the company's own employees. The system's track record and model graduation features are there to check and retrain them on the company's history as it builds up.
- **Promotion rewards improvement.** In the reference data, employees whose score did not rise were rarely promoted, so a consistently high scorer with no improvement shows as *Low*. The screen gives this as the reason.
- **Forecasts drift up about 2 points a year,** following the trend in the reference data.
- **Attrition rests on 155 survey answers.** Its 95% range is 0.547 to 0.731, and it did not carry over between the two survey groups. More responses are needed before it can be relied on.
- **"As run" results are a replay.** The previous models' system conditions were rebuilt from their code. Where a choice was needed (salary left out rather than sent in the wrong unit; the dataset's own employment types), it favoured the previous model.